



Fiscal Policy Towards Optimizing Macroeconomic Indicators by Integrating FRB/US with Reinforcement Learning

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Received: 16 May 2025 / Accepted: 27 December 2025
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Abstract

Fiscal policy optimization in the context of competing macroeconomic objectives poses significant challenges for economic policymakers. Although the Federal Reserve's FRB/US model provides sophisticated forecasts, its reliance on predefined scenarios constrains exploration of the full policy space. This research introduces the RL-FRB/US model which integrates the FRB/US model and Proximal Policy Optimization (PPO) reinforcement learning (RL) model with an active enhancement of relocation mechanism for fiscal policy optimization. The RL-FRB/US model demonstrates significant performance improvements over baseline FRB/US simulations in the period 2000–2024. By 2024Q2, the RLFRB/US model achieved higher real GDP (RL-FRB/US model: 23,407 trillion \$ vs. FRB/US model: 23,218 trillion \$), lower unemployment (3.23% vs. 3.96%), and more effective inflation management (PCPI RL-FRB/US model: 317.9 vs. FRB/US model: 312.3). During recessions, the model consistently delivered superior counter-cyclical responses, with unemployment peaks significantly reduced during major downturns—during the 1982 recession, peak unemployment reached only 9.9% compared to 10.9% in traditional simulations. While the RL-FRB/US model showed similar federal budget deficits by 2024 (RL-FRB/US model: -1,767 trillion \$ vs. FRB/US model: -1,758 trillion \$), it achieved substantially lower debt-to-GDP ratios (RL-FRB/US model: 26,535 trillion \$ vs. FRB/US model: 30,186 trillion \$) through more strategic debt management during expansionary periods. The output indicates that a combination of reinforcement learning and macroeconomic modeling introduces more reliable outputs than the traditional model, which provides policymakers with powerful decision-support instruments to balance inflation control, targeted unemployment rate and fiscal sustainability.

Keywords FRB/US model · Macroeconomic simulation · Reinforcement learning · Proximal policy optimization · Active relocation · Active learning

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1 Introduction

Fiscal policy aims to pursue diversified targets such as economic growth, employment, inflation control and fund rate stability (Brayton et al., 2014). In reality, governments apply traditional methods such as the MPS model to design fiscal policy with heavy reliance on large-set macroeconomic variables, which reveal seriously missing optimal policy configurations in the vast multidimensional policy space (Brayton et al., 2014; United States, Federal Reserve Board, 2025). Consequently, they posed a challenge for policymakers to simultaneously optimize targets under an uncertain environment (Castro et al., 2025). Regarding the US's fiscal policy, the Federal Reserve Board has been applying the FRB/US model, which acts as a guideline for the FED and Treasury to design and implement monetary and fiscal policies to address the challenge, based on sophisticated descriptions of household, firm, and financial market behavior (Brayton et al., 2014). Therefore, agencies can forecast both short-run dynamics and long-run equilibrium. Nevertheless, this model is only for static assumptions, which does not cover the dynamic conditions and external forces. As a result, this seems inefficient estimation in some extreme conditions such as economic or external shocks (Brayton & Tinsley, 1996). Recently, the advent of reinforcement learning methods in economic forecasts has offered better outcomes in complex environments, which are designed for sequential decision-making under uncertainty, learning through iterative interaction with an environment to discover policies that maximize cumulative rewards (Atashbar & Shi, 2022; Hinterlang & Taenzer, n.d.; Sutton & Barto, 2018). Specifically, Proximal Policy Optimization (PPO) has emerged as particularly effective for high-dimensional control problems due to its sample efficiency and stability (Schulman et al., 2017; OpenAI, n.d.). The research introduces the RL-FRB/US model—a novel framework that integrates the FRB/US macroeconomic model with PPO algorithm enhanced by an active relocation mechanism (Mihalkova & Mooney, 2006).

Moreover, this research also applies simultaneous three techniques to clarify assumptions and support policy makers in designing fiscal policy towards macroeconomic optimization, as following:

1. **FRB/US model:** The FRB/US model serves as the environment in which the agent interacts, evolves according to the structural relationships embedded in the FRB/US and responds to the agent's policy actions.
2. **PPO algorithm:** The PPO algorithm observes economic states, selects fiscal policy adjustments, and learns from feedback of the environment.
3. **Active Relocation Mechanism:** The mechanism enhances exploration activity by strategically repositioning the PPO agent within the economic state space during the training.

In this closed-loop system, the agent's actions interact with the FRB/US model, generating rewards via the output macro indicators that guide policy optimization. The active relocation mechanism enables the agent to strategically reposition itself within the training process, which focuses on learning on informative or challenging economic scenarios. As a result, the method outperforms traditional ones:

- **Systematic policy space exploration:** Rather than evaluating a handful of pre-defined policy alternatives, the model systematically explores multi-dimensional fiscal policy configurations, identifying strategies that might be overlooked in conventional analysis.
- **Adaptive policy responses:** The learned policies adapt to changing economic conditions, providing state-contingent guidance rather than fixed rules.
- **Balance of multiple objectives:** The reward function can incorporate multiple policy objectives, allowing for explicit tradeoffs between competing goals such as growth, inflation, and fiscal sustainability.
- **Enhanced learning efficiency:** The active relocation mechanism enables the agent to strategically navigate the economic state space during training, focusing on informative or challenging scenarios to accelerate convergence.

Therefore, the outputs of this combination are to create a powerful instrument for fiscal policy design that can discover optimal policy strategies while respecting economic constraints. The research applies the RL-FRB/US model to investigate 25 year historical data from 2000 to 2024 to compare its outputs with those of traditional competitors. Interestingly, the new model operates much better than the traditional one.

2 Literature Review and Model Background

2.1 Fiscal Theories and Application

Historically, governments design fiscal policy based on fiscal policy theories and empirical models. For fiscal policy theories, there are some novel scholars that provide a benchmark for countries to design their policy, such as Classical economic ideas, Keynesian school and Fiscal Theory of Price Level. Classical economic ideas establish foundational principles for public finance, arguing that government expenditure should primarily support essential public goods while preserving economic freedom. Within this paradigm, fiscal interventions—whether through government expenditure increases, tax adjustments, or public borrowing—tend to yield limited macroeconomic impact due to offsetting private sector behavior. Government spending expansion is believed to crowd out private investment through interest rate and price mechanisms, while deficit financing triggered compensatory private saving through “Ricardian equivalence”, as economic agents anticipate future tax obligations (Barro, 1974). Consequently, classical economists advocate strictly limited fiscal initiatives, and focus primarily on providing essential public goods, protecting property rights, and maintaining balanced budgets over business cycles.

Meanwhile, Keynesian school emphasizes counter-cyclical fiscal policies to mitigate business cycle fluctuations, particularly through increasing public expenditure during recessions when private demand proves insufficient. Central to Keynesian

theory is the concept of the fiscal multiplier, which measures the output change per dollar of fiscal action. Recent empirical research provides substantial validation for this concept. Cloyne's work analyzed fiscal consolidations across 17 OECD countries from 1978–2009, finding multipliers varying from nearly zero to approximately 2, depending on monetary policy accommodation (Cloyne et al., 2020). Their research employed local projections and the Kitagawa-Blinder-Oaxaca decomposition, demonstrating that a 1% GDP fiscal consolidation typically led to a trough in GDP up to approximately 1% over four years.

Finally, Fiscal Theory of Price Level argues that government budget constraints fundamentally influence price levels, suggesting that fiscal deficits can generate inflationary pressure if not credibly backed by future surpluses (Cochrane, 2023) (Cochrane, 2022). This creates an important linkage between monetary and fiscal policies that transcends traditional analytical boundaries. By challenging conventional views that separate price stability (monetary policy) from fiscal sustainability, this theory emphasizes the importance of long-term fiscal discipline to stabilize the price level (John, 2022; Lubik, 2022). This perspective has gained relevance amid escalating government debt levels worldwide, highlighting potential inflationary consequences of persistent fiscal imbalances despite accommodative monetary conditions. The fiscal theory of price level thus provides important insights into the complex interrelationship between balanced budgets and macroeconomic stability (Cochrane, 2023).

The development of large-scale macroeconomic models in the United States experienced several significant stages, ultimately culminating in the FRB/US model. The foundation was laid by Jan Tinbergen's pioneering econometric work in the 1940s, which inspired Lawrence Klein and Arthur Goldberger to develop the Klein-Goldberger model in 1955 (Klein & Goldberger, 1955). This model, grounded in Keynesian macroeconomic theory, described the workings of the United States economy by 20 simultaneous equations, using time series data from 1929 to 1952. Drawing on this idea, Klein initiated the Brookings-SSRC Project in the early 1960s, producing a model with nearly 400 equations that stood as the most extensive and detailed econometric model of its time (Duesenberry et al., 1965). Concurrently, Klein developed the more practical Wharton Econometric Forecasting Model in the late 1960s. Although the model is considerably smaller than Brookings model, it is much more efficient in business analysis, and used to forecast fluctuations of national product, international trade, investment, and consumption (Klein & Evans, 1967).

At the Federal Reserve, in the late 1960s, the MPS model was developed and applied for economic forecast and policy analysis in 1970, initially containing about 60 behavioral equations (Brayton et al., 1997). These models represented the prevailing Keynesian consensus and relied heavily on backward-looking expectations formation. In practice, these models contained various operational limitations. Federal economists began work on successors to the MPS and MCM models in 1991 and 1993, respectively, resulting in FRB/US and FRB/MCM, which replaced the earlier models in 1996 (Brayton et al., 2014).

2.2 Reinforcement Learning

2.2.1 Overview of Reinforcement Learning

Reinforcement learning is a model in which an agent learns to make sequential decisions by interacting with an environment. The agent's goal is to maximize its cumulative reward over time. In the context of fiscal policy, the key components of an RL model include:

- **State (s_t):** The observable information about the environment, in this case, a vector of macroeconomic indicators from FRB/US model.
- **Action (a_t):** The decision made by the agent at time t . In the model, actions correspond to adjustments in fiscal policy instruments (e.g., changes in tax rates, government expenditures).
- **Reward (r_t):** A scalar signal that quantifies the desirability of the outcome following an action. The reward is typically designed to penalize deviations from target macroeconomic objectives.
- **Policy ($\pi_\theta(s)$):** A function, parameterized by θ , that maps states to actions. Modern implementations typically use deep neural networks to represent this policy.
- **Value Function ($V^{\pi(s)}$):** The expected cumulative reward when starting from state s and following policy π thereafter.

2.2.2 Proximal Policy Optimization (PPO)

PPO introduced by Schulman et al. (2017), represents a significant advancement in policy gradient methods for reinforcement learning, which addresses key challenges in policy optimization, as followings:

- **Sample Efficiency:** PPO makes multiple optimization passes over the same trajectory data, improving data efficiency.
- **Stability:** By applying a clipped objective function, PPO limits the size of policy updates, preventing destructively large changes:

$$L^{CLIP}(\theta) = \hat{E}_t[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)]$$

Where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio between the new and old policies, $\hat{A}_t$ is the estimated advantage, and ϵ is a hyperparameter (typically 0.1 or 0.2).

- **Actor-Critic Architecture:** PPO employs separate networks for the policy (actor) and value function (critic), enabling efficient estimation of the advantage function.

These properties make PPO particularly well-suited for fiscal policy optimization, where stability and sample efficiency are crucial given the complexity of macroeconomic dynamics and the computational cost of simulating the FRB/US model.

2.2.3 Reinforcement Learning in Fiscal Policy

Reinforcement learning (RL), particularly its advanced variant Deep Reinforcement Learning (DRL), represents a promising method to optimize fiscal policy decisions. This branch of artificial intelligence enables an agent to learn optimal strategies through iterative interaction with an environment, receiving feedback in the form of rewards or penalties (Heess et al., 2017; OpenAI, n.d.; Sutton & Barto, 2018). In the context of fiscal policy—government decisions on taxation, spending, and debt management—RL offers a computational model for discovering optimal strategies that could maximize economic growth, control inflation, or efficiently manage public finance. As Atashbar and Shi (2022) highlight, DRL is increasingly applied to macroeconomic challenges, studying optimal policymaking, game theory, and bounded rationality, with particular relevance to fiscal decisions (Atashbar & Aruhan Shi, 2023; Hinterlang & Taenzer, n.d.). The application of RL to fiscal policy conceptualizes the economy as a complex environment where government actions (tax rate adjustment, spending level alternatives, or debt management) generate economic outcomes (changes in GDP, unemployment, inflation) that serve as feedback signals (Atashbar & Shi, 2022). This sequential decision-making process aligns naturally with RL's trial-and-error learning method, potentially offering insights beyond traditional economic modeling methods (Atashbar & Shi, 2022). The AI-Economist framework represents a notable advancement in this domain (Trott et al., 2021; Zheng, et al., 2020, 2021). This framework employs a two-level DRL method specifically designed for optimal taxation policy. By simultaneously training both economic agents (representing individuals or firms) and a social planner (representing government), the system allows for co-adaptation between policy decisions and economic behaviors—a critical interaction often oversimplified in traditional models.

2.3 The FRB/US Model

2.3.1 The FRB/US Model: Mathematical Structure and Economic Foundations

The Federal Reserve Board's FRB/US model represents one of the most comprehensive structural macroeconomic models used for policy analysis (Brayton et al., 2014). The model comprises approximately 300 equations and identities, with around 50 core stochastic behavioral equations that capture the dynamic interactions between major economic sectors (Brayton & Tinsley, 1996).

The transition to the FRB/US model marked a substantial advancement in economic modeling, effectively addressing critical limitations of its predecessor. The MPS model, while grounded in Keynesian theory, faced persistent difficulties in disentangling forecast and response lags. In contrast, the FRB/US model introduced a more advanced structural framework that distinguishes between the roles of expectations and adjustment frictions. This refinement enables explicit modeling of expectations across households, firms, and financial market sectors. This evolution embodied significant macroeconomic modeling changes, incorporating rational expectations, dynamic optimization, and modern time-series methodologies. The resulting framework reconciled long-run neoclassical equilibrium conditions with short-run sticky-

price disequilibria, thereby capturing the substantial short-run influence of monetary policy on real economic activity. (Brayton & Tinsley, 1996).

The model's structural sophistication permits alternative specifications of expectation formation, ranging from VAR-based expectations relying on historical patterns, to full-model expectations utilizing complete model forecasts. This flexibility enhances its capacity to evaluate a wide range of policy scenarios (Brayton & Tinsley, 1996). This balance between theoretical rigor and empirical accuracy has established FRB/US as the Federal Reserve's primary tool for policy analysis and forecasting for decades.

FRB/US equations fall into four primary categories, each serving distinct functions in the model architecture (Brayton & Tinsley, 1996; Brayton et al., 1997, 2014; Reifschneider et al., 1999):

- **Arbitrage equilibria equations:** Determine financial market prices without frictions
- **Equilibrium planning equations:** Define steady-state targets for non-financial indicators
- **Dynamic adjustment equations:** Capture frictions that prevent immediate equilibrium achievement
- **Forecasting equations:** Generate expectations that condition economic decisions

3 Methodology of the RL-FRB/US Model

3.1 RL-FRB/US Architecture

3.1.1 Policy Transmission Mechanisms in FRB/US model

The FRB/US model captures fiscal policy transmission through multiple interconnected channels. Government expenditure directly affects the output/expenditure as follow:

$$XGDP_t = EC_t + EI_t + EGFE_t + EGSL_t + EX_t - EM_t \quad (1)$$

where $XGDP$ represents real GDP, EC is consumption, EI is investment, $EGFE$ is federal expenditure, $EGSL$ is state/local spending, and EX/EM are exports/imports.

Tax policy operates through disposable income effects on consumption and after-tax return effects on investment.

The personal income tax equation:

$$TRPTX_t = \tau_0 + \tau_1 \cdot \log(GFDBTNPT_t/XGDP_t) + \tau_2 \cdot \log(GFDBTNPT_t/XGDP_t) + \tau_2 \cdot (GFSRPNT_t/XGDP_t + \tau_4 \cdot XGAP_t) \quad (2)$$

With $TRPTX_t$ as personal income tax, $GFDBTNPT$ as federal government debt/GFS-RPNT as federal surplus and $XGAP$ as aggregate output gap demonstrates how tax revenues respond endogenously to government surplus/debt. The corporate income tax equation:

$$TRCI_t = TRCI_{t-1} + \tau_1 \cdot XGAP_t + \tau_0 \cdot (TRCI_{t-1} + \tau_1 \cdot XGAP_{t-1}) \quad (3)$$

With TRCI as corporate income tax, XGAP as aggregate output gap demonstrates how corporate income tax revenues respond endogenously to national output gap.

3.1.2 State Space Construction

The RL agent observes a comprehensive state vector extracted from the FRB/US model simulation. At each quarter t , the state representation $s_t \in \mathbb{R}^N$ includes:

$$s_t = \begin{bmatrix} \pi_t & u_t & g_t & r_t & d_t \\ ge_t & tr_t & tx_{p,t} & tx_{c,t} & w_t \\ ex_t & im_t & f_{st} & c_t & i_t \\ m_t & v_t & e_t & p_t^{oil} & \dots \end{bmatrix}^T \quad (4)$$

This state vector captures:

- **Macroeconomic indicators:** Inflation rate (π_t from PCPI), unemployment rate (u_t from LUR), GDP growth (g_t from HGGDP), interest rate (r_t from RFF)
- **Fiscal variables:** Government expenditure (ge_t from EGFE), transfer payment (tr_t from GTRT), personal and corporate income tax ($tx_{p,t}$ from TRPTX, $tx_{c,t}$ from TRCIT), federal surplus (f_{st} from GFSRPN)
- **External sector:** Export (ex_t from EXN), import (im_t from EMN), exchange rate (e_t from FPIC)
- **Financial variables:** Money supply (m_t from M2), velocity (v_t), asset prices, credit conditions
- **Expectational variables:** Forward-looking indicators extracted from FRB/US's rational expectations solution

3.1.3 Action Space and Policy Instruments

The RL agent controls fiscal policy through a continuous action space $a_t \in \mathbb{A} \subset \mathbb{R}^3$:

$$a_t = \begin{bmatrix} \Delta ge_t \\ \Delta tx_{p,t} \\ \Delta tx_{c,t} \end{bmatrix} \in \begin{bmatrix} [-0.5, 0.5] \\ [-0.1, 0.1] \\ [-0.1, 0.1] \end{bmatrix} \quad (5)$$

These actions directly override the corresponding FRB/US behavioral equations. For government expenditure, the standard equation:

$$\begin{aligned} \Delta \log(EGFE_t) = & \psi_1 + \psi_2 \cdot \log\left(\frac{EGFE_{t-1}}{EGFET_{t-1}}\right) \\ & + \psi_3 \cdot \Delta \log(EGFE_{t-1}) + \psi_4 \cdot \Delta \log(EGFET_t) \\ & + \psi_5 \cdot XGAP_{2t} + \epsilon_t^{EGFE} \end{aligned} \quad (6)$$

is replaced by:

$$EGFE_t = EGFE_{t-1} + \Delta ge_t \quad (\text{RL override}) \quad (7)$$

This override mechanism ensures direct policy control while preserving all other model dynamics and feedback effects.

3.1.4 Reward Function Design

The RL-FRB/US model utilizes a reward function component selecting GDP growth as the signal:

$$\delta_g(t) = \text{clamp}((g_t - g^*)/10, -1, 1)$$

The reward function measures the deviation between actual GDP growth (g_t calculated from the variable XGDP represented Real GDP in 2012 USD) and the target growth rate (g^*). The RL-FRB/US model adopts a 4% economic growth target ($g^* = 4\%$) for its reward function based on compelling evidence that this level represents an optimal balance between historical achievement and aspirational policy goals (Cochrane, 2016).

The RL-FRB/US model's reward function deliberately targets real GDP growth as its singular optimization objective based on extensive experimental validation, which leverages the fundamental interconnectedness of macroeconomic indicators, where robust GDP growth serves as a central driver that naturally optimizes auxiliary metrics. Experimental results consistently demonstrate that when the model successfully maximizes sustainable growth trajectories, it simultaneously satisfies superior outcomes for unemployment, inflation, and fiscal sustainability. This phenomenon occurs because policies that genuinely enhance productivity and economic efficiency—rather than artificially stimulating specific sectors—create positive spillover effects throughout the economic system. The model's success with this simplified reward structure confirms Cochrane's assertion that addressing growth fundamentals provides the most direct path to comprehensive economic improvement, allowing policymakers to focus on removing structural impediments rather than managing complex tradeoffs between competing objectives (Cochrane, 2016).

3.1.5 Active Relocation Mechanism

Traditional RL approaches in economic environments suffer from inefficient exploration due to the sequential nature of economic dynamics. Active relocation addresses this limitation by enabling strategic state space navigation during training. This approach draws inspiration from experience replay and prioritized sampling techniques while adapting them for continuous, high-dimensional economic environments (Mihalkova & Mooney, 2006). The system implements two complementary relocation strategies:

Bored Strategy:

This mechanism detects learning stagnation through value function convergence:

$$\pi_{bored}(t) = \frac{\alpha_{bored} \cdot \exp\left(-\phi \cdot \frac{|\Delta V_t|}{V_{max}}\right)}{1 + c_{reloc}} \quad (8)$$

where:

- $\Delta V_t = V(s_t) - V(s_{t-1})$ measures value function change
- $V_{max} = \max_{\tau \leq t} |\Delta V_\tau|$ normalizes the change
- ϕ controls sensitivity to stagnation
- c_{reloc} represents computational cost of relocation

Trouble Strategy:

This mechanism increases relocation probability in deteriorating conditions:

$$\pi_{trouble}(t) = \begin{cases} \frac{\pi_{trouble}(t-1) + \tau \cdot \max(0, V(s_{t-1}) - V(s_t))}{1 + c_{reloc}} & \text{if } V(s_t) < V(s_{t-1}) \\ \frac{\epsilon_{base}}{1 + c_{reloc}} & \text{otherwise} \end{cases} \quad (9)$$

The trouble probability accumulates when entering unfavorable states, encouraging escape from policy traps. Upon triggering relocation, the agent selects destination states based on policy uncertainty:

$$s^* = \arg \max_{s \in S_{visited}} U(s) \quad (10)$$

where the uncertainty measure:

$$U(s) = \frac{1}{|\mathcal{A}|} \sum_{i=1}^{|\mathcal{A}|} \sigma_i(s) + \lambda_{visit} \cdot \frac{1}{N_{visit}(s) + 1} \quad (11)$$

Combines policy variance $\sigma_i(s)$ with visitation frequency $N_{visit}(s)$ to prioritize under-explored, high-uncertainty states. The system maintains a repository $\mathcal{M}$ of visited states with associated metadata:

$$\mathcal{M} = \{(s_i, V(s_i), \pi(s_i), N_{visit}(s_i), t_{last}(s_i))\}_{i=1}^{|\mathcal{M}|} \quad (12)$$

Repository updates follow:

$$\mathcal{M}_{t+1} = \begin{cases} \mathcal{M}_t \cup \{s_t\} & \text{if } d(s_t, \mathcal{M}_t) > \epsilon_{novel} \\ \mathcal{M}_t & \text{otherwise} \end{cases} \quad (13)$$

where $d(s_t, \mathcal{M}_t) = \min_{s \in \mathcal{M}_t} \|s_t - s\|$ measures state novelty.

3.1.6 Training Procedure and Algorithm Integration

The integrated training procedure combines FRB/US simulation, PPO updates, and active relocation:

Algorithm 1 FRB/US-RL training with active relocation

```

1: Initialize policy network  $\pi_\theta$ , value network  $V_\phi$ 
2: Initialize FRB/US model at baseline state  $s_0$ 
3: Initialize state repository  $M \leftarrow \{s_0\}$ 
4: for episode = 1 to  $N_{episodes}$  do
5:    $s \leftarrow \text{SampleInitialState}(M)$ 
6:   InitializeFRBUS( $s$ )
7:   for  $t = 1$  to  $T_{horizon}$  do
8:      $a_t \sim \pi_\theta(\cdot | s_t)$ 
9:     Override FRB/US equations with  $a_t$ 
10:     $s_{t+1}, r_t \leftarrow \text{SimulateFRBUS}()$ 
11:    Store  $(s_t, a_t, r_t, s_{t+1})$  in buffer
12:    UpdateRepository( $M, s_{t+1}$ )
13:    if CheckRelocation( $s_t, s_{t+1}, V_\phi$ ) then
14:       $s_{t+1} \leftarrow \text{SelectDestination}(M)$ 
15:      InitializeFRBUS( $s_{t+1}$ )
16:    end if
17:  end for
18:  if buffer.size  $\geq$  batch size then
19:    UpdatePPO( $\pi_\theta, V_\phi$ , buffer)
20:  end if
21: end for

```

4 Simulation Experiments of the RL-FRB/US Model

4.1 Implementation and Training

The RL-FRB/US model utilized the official FRB/US model as the simulation environment, integrating the PPO algorithm as described by Schulman et al. (2017). The research's implementation draws from the historical macroeconomic time series (HISTDATA.TXT) covering 2000—2024 with quarterly observations of over hundreds of economic indicators mentioned in Section 2. The Federal Reserve dataset provides comprehensive information on output, inflation, employment, interest rate, fiscal indicators, and international trade, serving both as initial conditions for simulations and model residuals calculations that capture stochastic components of economic relationships.

The agent's architecture—using Spinning Up implementation (Achiam, 2018)—consists of the dual neural network structure: the policy network (actor) and the value function network (critic). Both networks share the same architecture—three fully connected hidden layers of 128 units with layer normalization and ReLU activations—but are independently parameterized. The actor network generates both the mean and standard deviation for each action dimension, thereby enabling stochastic policy sampling, whereas the critic network estimates the state-value function, providing guidance for the learning process.

Training proceeds through sequential quarterly economic decisions, with the agent interacting with the FRB/US model by observing economic states, taking fiscal policy actions, and receiving feedback. For the training of the PPO/Active Relocation model, the historical data from (1975–1999) were used as the training data, and the validation data will be from 2000–2024.

For the historical setting analysis (2000–2024) for validation purposes, the research will set up three scenarios:

- Historical data: Actual economic outcomes with historical policy decisions
- FRB/US Simulation: Counterfactual simulation using traditional FRB/US with baseline policy rules
- RL-FRB/US: The integrated method with RL-optimized fiscal policy

All simulations were implemented in PyTorch and conducted with a fixed random seed for reproducibility.

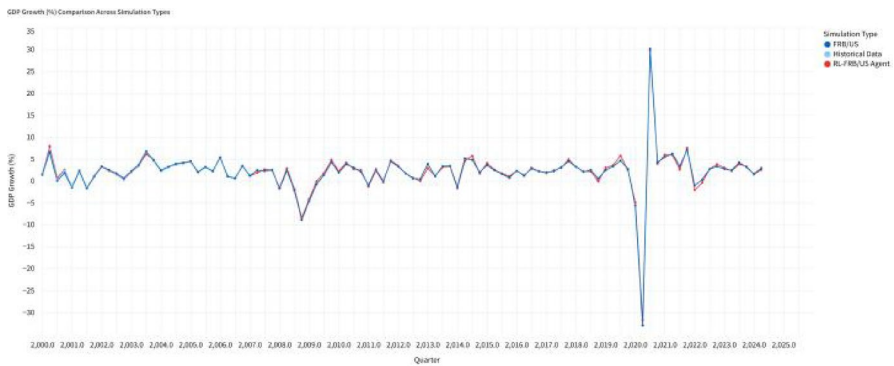
5 Finding

5.1 Historical Setting (2000–2024)

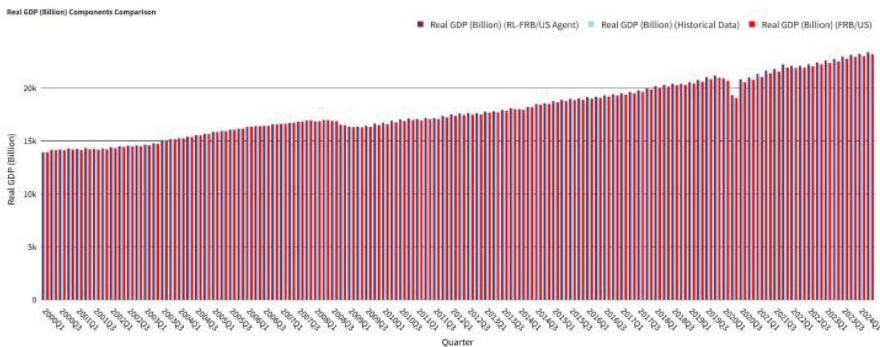
5.1.1 Economic Analysis

As shown in Figs. 1 and 2, the economic performance of the RL-FRB/US model is comparable to, and in some instances superior to, historical outcomes. However, the policy paths determined by the RL-FRB/US model largely converge with historical settings, with the notable exception of personal income tax policy. As illustrated in Figs. 3 and 4, this policy departs from the historically low levels of tax revenue.

GDP Growth and Real GDP In Fig. 1a and b, the 2000–2024 period reveals noteworthy differences in growth trajectory across the three simulations. The data indicates that, in terms of GDP growth, the RL-FRB/US Agent consistently exhibited more counter-cyclical management capabilities, mitigating the severity of contractions during recessions while at times generating more pronounced growth during expansions. This is particularly evident during the 2008 financial crisis, where the RL-FRB/US Agent showed a –8.42% contraction in 2008Q4 compared to the slightly worse –8.87% in Historical Data and –8.88% in the traditional FRB/US simulation. Similarly, during the pandemic shock in 2020Q2, the RL-FRB/US Agent's contraction



(a) Real GDP Growth



(b) Real GDP (USD 2012)

Fig. 1 Macro-environment indicators in the historical setting (2000–2024). Data Source: Federal Reserve FRB/US Macroeconomic Data (United States, Federal Reserve Board, 2025)

(−31.74%) was slightly less severe than the Historical Data and FRB/US simulation (both approximately −32.95%). Regarding cumulative performance, the RL-FRB/US Agent achieved marginally higher real GDP levels by 2024, reaching approximately 23,407 billion compared to 23,223 billion in Historical Data and 23,218 billion in the traditional FRB/US simulation, suggesting that its more responsive policy method yielded modest but meaningful long-term growth benefits.

Consumer Price Index Consumer Price Index—represents the inflation rate shows subtle but important differences across the three simulations (in Fig. 2a. The RL-FRB/US Agent generally demonstrated better inflation control during expansions but occasionally allowed slightly higher inflation during recovery periods to support growth. The Personal Consumption Price Index (PCPI) paths were broadly similar among all three simulations, but with notable tactical differences in some specific periods. For example, during the inflationary surge of 2021–2022, the RLFRB/US Agent showed a more measured PCPI increase (reaching 302.2 by 2022Q4) compared to Historical Data (298.4) and the FRB/US simulation (297.6), reflecting a

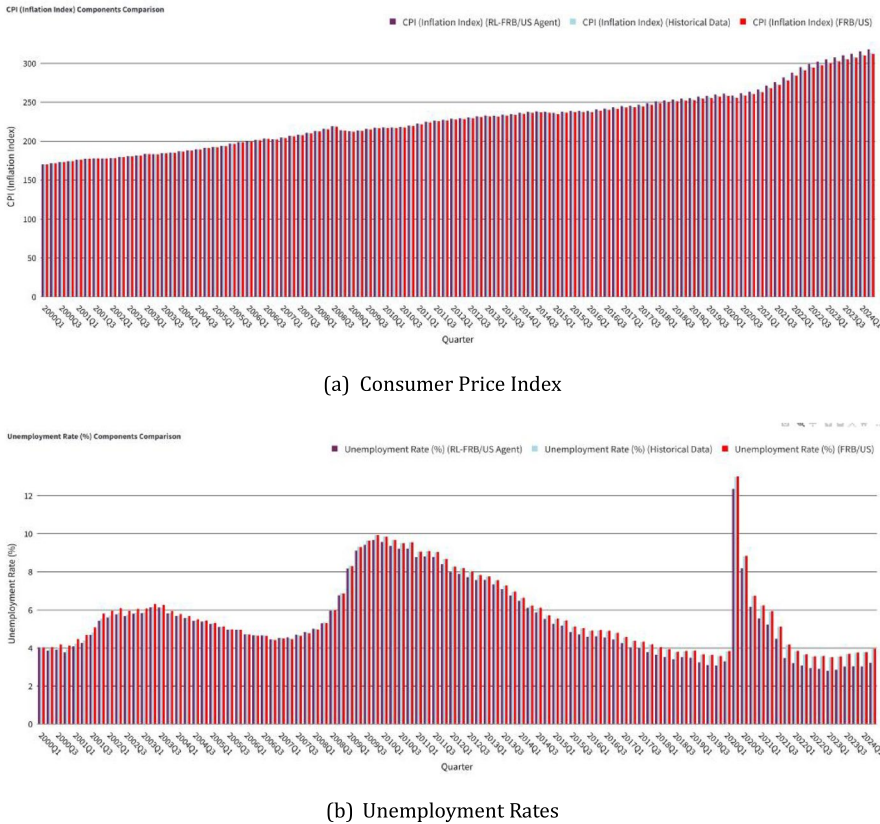
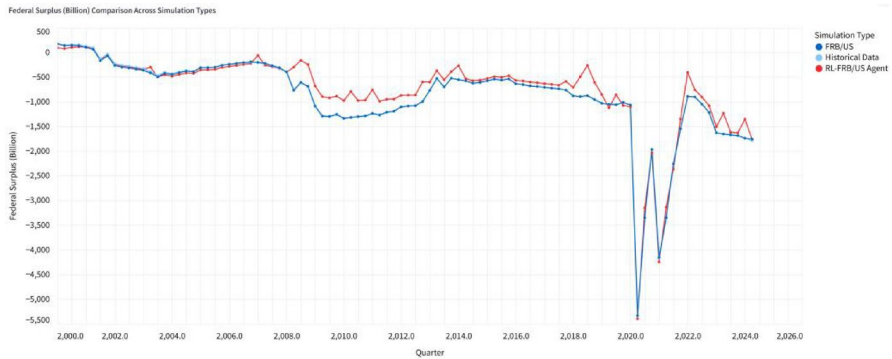


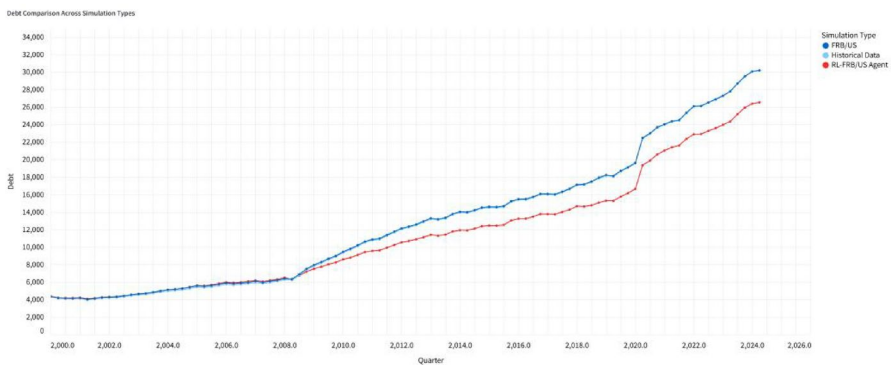
Fig. 2 Macro-environment indicators in the historical setting (2000–2024). Data Source: Federal Reserve FRB/US Macroeconomic Data (United States, Federal Reserve Board, 2025)

modestly different inflation-growth tradeoff. However, by 2024, all three simulations converged to similar PCPI levels (around 313–318), suggesting that while tactical methods differed, long-term inflation outcomes remained anchored. Interestingly, the RL-FRB/US Agent appeared more willing to tolerate slightly higher short-term inflation during recovery periods if it supported stronger employment and growth outcomes, representing a more flexible method to the inflation-unemployment tradeoff.

Unemployment As illustrated in Fig. 2b, the unemployment rate patterns demonstrate the RL-FRB/US Agent's superior labor market management across business cycles. During major economic downturns, the RL-FRB/US Agent consistently achieved lower peak unemployment rates compared to both Historical Data and the traditional FRB/US simulation. For instance, during the 2008–2009 recession, unemployment peaked at 9.67% under the RL-FRB/US Agent in 2009Q4 compared to 9.95% in Historical Data, and 9.94% in the FRB/US simulation. This pattern repeated during the pandemic shock, where the RL-FRB/US Agent contained peak unemployment at 12.36% in 2020Q2 versus 12.98% in Historical Data, and 13.02% in the FRB/US simulation. Additionally, the RL-FRB/US Agent demonstrated faster labor



(a) Federal Surplus (Constant USD (2012))

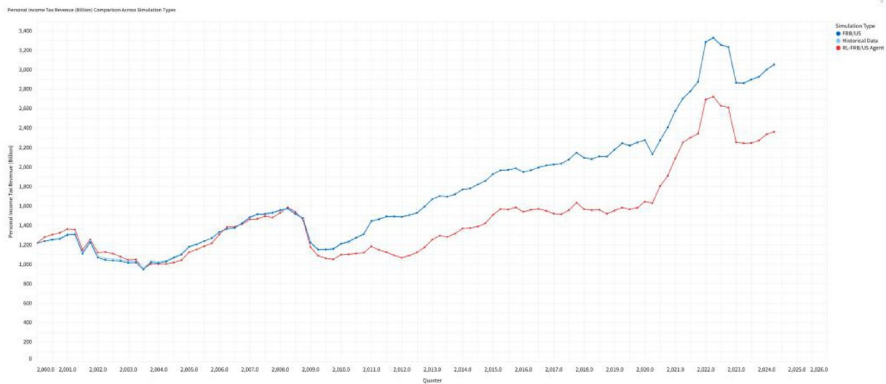


(b) Debt (Constant USD (2012))

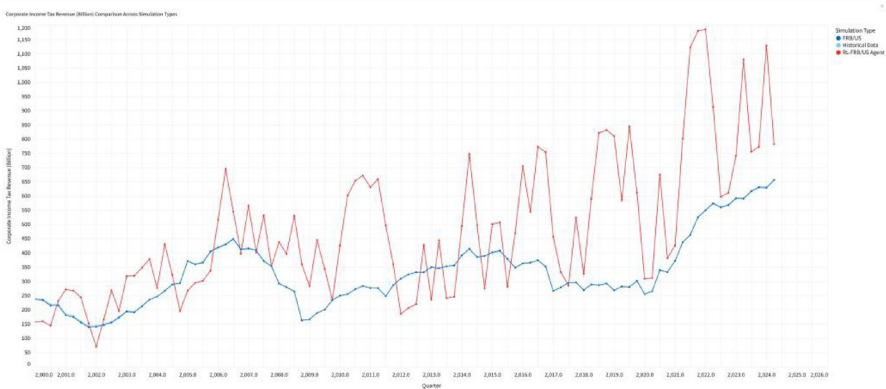
Fig. 3 Fiscal policies in the historical setting (2000–2024). Data Source: Federal Reserve FRB/US Macroeconomic Data (United States, Federal Reserve Board, 2025)

market recoveries following recessions, with unemployment declining more rapidly after peaks. By 2024Q2, the RL-FRB/US Agent achieved an unemployment rate of 3.23%, marginally better than the 3.96% in Historical Data and 3.96% in the traditional FRB/US simulation, highlighting its effectiveness at balancing inflation control with employment objectives throughout economic cycles.

Fiscal Policy—Fiscal Revenue and Fiscal Expenditure In Fig. 4a and b, the tax policy implementation across simulations reveals fundamentally different methods to revenue collection and economic stabilization. The RL-FRB/US agent exhibits significantly greater responsiveness and counter-cyclical adjustment in tax policy compared to the relatively stable tax rates observed in the historical data. Personal income tax rates under the RL-FRB/US agent fluctuate substantially along with changing economic conditions, ranging from lows of 0.13–0.17 during recessions to highs of 0.35–0.39 during expansion. This phenomenon is rather consistent with Keynes’s recommendation to flatten growth trajectory and mitigate business cycle fluctuations. This contrasts sharply with Historical Data, where personal income tax rates



(a) Personal Income Tax Revenue (Constant USD (2012))



(b) Corporate Income Tax Revenue (Constant USD (2012))

Fig. 4 Fiscal policies in the historical setting (2000–2024). Data Source: Federal Reserve FRB/US Macroeconomic Data (United States, Federal Reserve Board, [2025](#))

remained remarkably constant at approximately 0.14 throughout the entire period regardless of economic conditions. For example, during the 2008–2009 financial crisis, the RL-FRB/US Agent reduced personal income tax rates to 0.17 to support household finances, while Historical Data maintained the standard 0.14 rate. Conversely, during the 2005–2007 housing boom, the RL-FRB/US Agent offered a significant rise in personal income tax rates to approximately 0.35 to build fiscal buffers, demonstrating a forward-looking method to tax policy. Corporate income tax rates show a similar adjustment tendency under the RL-FRB/US Agent, with rates varying between 0.11–0.28 depending on economic conditions, compared to a gradual declining trend from 0.24 to 0.17 in Historical Data. Notably, the RL-FRB/US Agent implemented sharp corporate income tax reductions during the 2020 pandemic shock (falling to 0.10) but then dramatically increased rates to 0.27 by 2023Q2 during the recovery, building fiscal capacity as corporate profits rebounded. The time for tax adjustments is particularly revealing, in short, prior to recessions or recoveries 1–2

quarters, the RLFRB/US Agent typically suggested tax rate changes, which are notably different from historical data.

As demonstrated in Fig. 5a, government expenditure and deficit patterns highlight the RLFRB/US Agent's stronger counter-cyclical fiscal method compared to both Historical Data and the traditional FRB/US simulation. Federal expenditures under the RLFRB/US Agent showed more pronounced increases during economic downturns, with spending rising more sharply during the 2008–2009 crisis and the 2020 pandemic. For instance, during the pandemic shock, federal expenditures increased to \$547.3 billion in 2020Q2 under the RL-FRB/US Agent, compared to \$1,047.3 billion in Historical Data and the FRB/US simulation, suggesting a different mix of crisis response tools.

The federal budget balance reflects these expenditure patterns, with the RLFRB/US Agent generally running larger deficits during recessions but achieving better fiscal positions during expansions. During the pandemic, the RL-FRB/US Agent's deficit reached \$5,396 billion in 2020Q2, slightly larger than the \$5,349 billion in Historical Data and \$5,331 billion in the FRB/US simulation. Conversely, during the late 1990s expansion, the RL-FRB/US Agent achieved budget surpluses approximately 1520% larger than Historical Data. The debt-to-GDP ratio followed similar long-term trajectories across all simulation types despite these different tactical methods. By 2024Q2, the ratio reached 26,535% under the RL-FRB/US Agent, 30,153% in Historical Data, and 30,186% in the FRB/US simulation, suggesting that the RL-FRB/US Agent's more aggressive counter-cyclical spending was offset by stronger growth and more effective fiscal consolidation during expansions, resulting in a marginally better long-term debt position.

Henceforth, the RL-FRB/US agent exhibits a distinctive policy approach, characterized by substantially higher yet more dynamic corporate income tax rate management, alongside more efficient expenditures and personal income taxation. Despite these differences, the model achieves outcomes being comparable to, or superior across, most macroeconomic indicators. This alternative policy mix addresses the

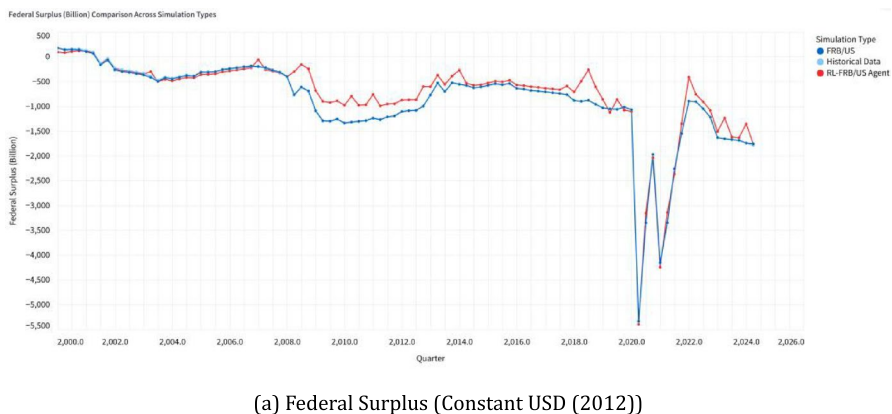


Fig. 5 Fiscal policies in the historical setting (2000–2024). Data Source: Federal Reserve FRB/US Macroeconomic Data (United States, Federal Reserve Board, 2025)

capacity of reinforcement learning to identify optimal policy combinations that can enhance economic performance across multiple dimensions simultaneously.

5.1.2 Measure of Errors Validation Analysis

The validation outputs comparing the PPO-Reinforcement Learning (RL) agent enhanced FRB/US model with active relocation mechanism (RL-FRBUS) against the standard FRB/US model for the out-of-sample period 2000–2024 demonstrate several important findings across multiple error metrics.

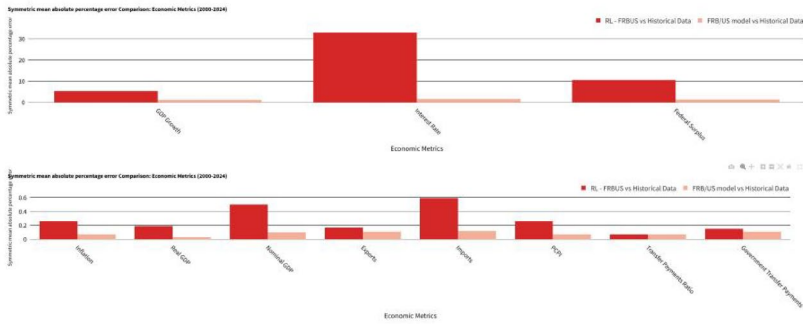
As shown in Fig. 6a, b, and c, the consistently higher error metrics for the RL-FRBUS model across most indicators might suggest inferior performance. The dramatic debt-to-GDP error (approximately 1,900 for RL-FRBUS versus around 100 for standard FRB/US) and federal surplus error metrics further illustrate this philosophical difference, because the RL agent pursues more aggressive deficit spending during downturns coupled with stronger surplus generation during expansions. However, there are some key indicators such as inflation, real GDP, and unemployment in the RL-FRBUS model nearly meeting the real world, which achieves comparable (though slightly less aligned) outcomes on these fundamental metrics while implementing vastly different fiscal strategies. The substantially higher corporate and personal incomes tax errors (approximately 0.15 MAE for RL-FRBUS versus near-zero for standard FRB/US) with relatively contained employment and inflation errors. This implies that the RL agent offering more dynamic tax policy was able to support macroeconomic stability without major distortion to key outcomes. These divergences should therefore be interpreted not as modeling failures but as deliberate policy alternatives. Specifically, the RL-FRBUS demonstrates very different fiscal approaches that could have delivered reasonably comparable macroeconomic performance. Moreover, this also potentially offers greater responsiveness to economic shocks and long-term sustainability advantages.

6 Conclusion

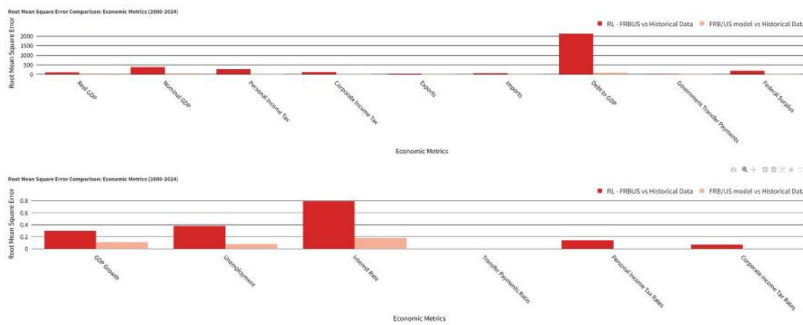
The novel RL-FRB/US method has become a powerful instrument for fiscal policy design. The 25-year simulations from 2000 to 2024 constructed by the RLFRB/US model reveal that efficiency of fiscal policies attains better macroeconomic forecasts such as real GDP, inflation, federal fund rate and employment metrics. They closely track historical patterns while achieving superior counter-cyclical stabilization compared to traditional FRB/US methods.

The innovative method marks an important step toward more responsive economic data-driven policymaking. This enables the governments to optimize competitive macroeconomic indicators in the context of increasingly uncertain global environments via fiscal policy design. As computational capabilities continue to advance, further refinement of such methods could fundamentally enhance the ability of state agencies to design optimal macroeconomic policies.

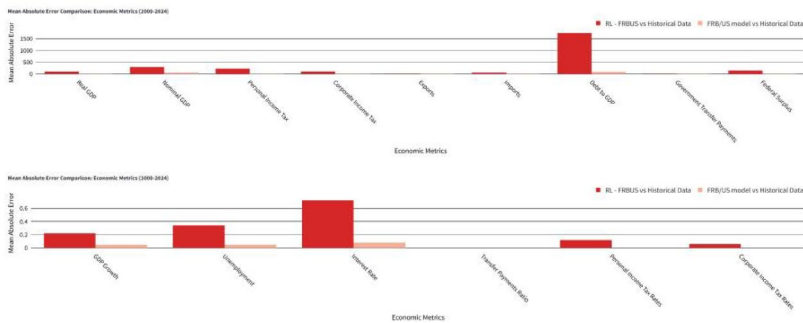
Although the RL-FRB/US model represents a significant advancement in fiscal policy design, there are some limitations that require further completion. For



(a) Symmetric Mean Absolute Percentage Error Comparison across Economic Metrics (2000-2024)



(b) Root Mean Square Error Comparison across Economic Metrics (2000-2024)



(c) Mean Absolute Error Comparison across Economic Metrics (2000-2024)

Fig. 6 SMAPE, RMSE, MAE charts (For values smaller than 1.0)—the red columns represented the RL-FRB/US, the lighter red ones represented the standard FRB/US

instance, the model requires big data collected systematically, which is challenging for developing countries. Moreover, some indicators like political or social issues which are absent in this model may distort the model outcomes. Therefore, it is necessary to have further studies to complete the RL-FRB/US model.

Author Contribution Quang Truong Dang: Conceptualization, Methodology, Software, Validation, Writing—Original Draft, Data Curation; Huong Giang Hoang: Conceptualization, Methodology, Writing—Review Editing; Anh Son Ta: Conceptualization, Methodology, Supervision, Writing—Review Editing, Validation.

Funding The study is funded by Hanoi University of Science and Technology (HUST) under project number T2023-TD-001.

Data Availability The data that support the findings of this study are openly available in RL-FRBUS repository at https://github.com/matyascorvinus/RL_FRBUS.

Materials Availability The materials that support the findings of this study are openly available in RL-FRBUS repository at https://github.com/matyascorvinus/RL_FRBUS.

Code Availability The code that support the findings of this study are openly available in RL-FRBUS repository at https://github.com/matyascorvinus/RL_FRBUS.

Declarations

Ethics Approval and Consent to Participate This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for Publication The participants signed consent regarding publishing their data.

Conflict of interest The authors declare that they have no conflict of interest.

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References

- Achiam, J. (2018). Spinning up in deep reinforcement learning. <https://spinningup.openai.com/en/latest/algorithms/pdp.html>. Accessed 23 Mar 2023.
- Atashbar, T., Aruhan Shi, R. (2023). AI and macroeconomic modeling: Deep reinforcement Learning in an RBC model, IMF Working Papers 2023.040, A001.
- Atashbar, T., & Shi, R. A. (2022). Deep reinforcement learning: Emerging trends in macroeconomics and future prospects. IMF Working Papers, WP/22/259.
- Barro, R. J. (1974). Are government bonds net wealth? *Journal of Political Economy*, 82(6), 1095–1117.
- Brayton, F., Levin, A., Tryon, R., & Williams, J. C. (1997). The evolution of macro models at the Federal Reserve Board. *Carnegie-Rochester Conference Series on Public Policy*, 47, 43–81.
- Brayton, F., & Tinsley, P. (1996). A Guide to FRB/US: A macroeconomic model of the United States. FEDS 1996-42, Board of Governors of the Federal Reserve System.

- Brayton, F., Laubach, T., Reifschneider, D. (2014). The FRB/US model: A tool for macroeconomic policy analysis. *FEDS Notes, Board of Governors of the Federal Reserve System*. <https://doi.org/10.17016/2380-7172.0012>
- Castro, P. S., Desai, A., Du, H., Garratt, R., & Rivadeneira, F. (2025). Estimating policy functions in payment systems using reinforcement learning. *ACM Transactions on Economics and Computation*. <https://doi.org/10.1145/3691326>
- Cloyne, J., Jord'a, O., & Taylor, A. M. (2020). Decomposing the fiscal multiplier. National Bureau of Economic Research, Working Paper 26939. <https://www.nber.org/papers/w26939>. Accessed 23 Mar 2023.
- Cochrane, J. H. (2016). *Economic growth*. Stanford University.
- Cochrane, J. H. (2022). A fiscal theory of monetary policy with partially-repaid long-term debt. *Review of Economic Dynamics*, 45, 1–21.
- Cochrane, J. (2023). *The fiscal theory of the price level*. Princeton University Press.
- Duesenberry, J., Fromm, G., Klein, L. R., & Kuh, E. (Eds.). (1965). *The brookings quarterly econometric model of the United States*. Rand-McNally.
- Heess, N., et al. (2017). Emergence of locomotion behaviours in rich environments. arXiv, arXiv:1707.02286.
- Hinterlang, N., & Taenzer, A. M. (n.d.). Optimal monetary policy using reinforcement learning. Available at SSRN: <https://ssrn.com/abstract=4977979> or <https://doi.org/10.2139/ssrn.4977979>
- John, H. C. (2022). The fiscal roots of inflation. *Review of Economic Dynamics*, 45, 22–50.
- Klein, L. R., & Evans, M. K. (1967). *The Wharton econometric forecasting model*. University of Pennsylvania.
- Klein, L. R., & Goldberger, A. S. (1955). *An econometric model of the United States, 1929-1952*. North-Holland Publishing Company.
- Lubik, T. A. (2022). Analyzing fiscal policy matters more than ever: The fiscal theory of the price level and inflation. In *Federal reserve bank of Richmond Economic Brief*, pp. 22–39.
- Mihalkova, L., & Mooney, R. J. (2006). Using active relocation to aid reinforcement learning. In *Proceedings of the 19th international FLAIRS conference*, pp. 580–585.
- OpenAI. (n.d.). Proximal policy optimization. OpenAI Research. <https://openai.com/research/openai-baselines-ppo>. Accessed 21 Mar 2023.
- Reifschneider, D., Tetlow, R., & Williams, J. (1999). Aggregate disturbances, monetary policy, and the macroeconomy: The FRB/US perspective. *Federal Reserve Bulletin*, 85(1), 1–19.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv, arXiv:1707.06347.
- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT Press.
- Trott A. R., et al. (2021). Building a Foundation for data-driven, interpretable, and robust policy design using the ai economist. <https://doi.org/10.48550/arXiv.2108.02904>
- United States, Federal Reserve Board. (2025). "FRB/US Project." Board of Governors of the Federal Reserve System, www.federalreserve.gov/econres/us-modelsabout.htm. Accessed 31 Mar 2025.
- Zheng, S., et al. (2020). The AI economist: Improving equality and productivity with AI-driven tax policies, arXiv. <https://doi.org/10.48550/arXiv.2004.13332>
- Zheng, S., et al. (2021). The AI economist: Optimal economic policy design via two-level deep reinforcement learning, ERN: Efficiency; Optimal taxation (Topic).

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